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Title:
MQTT Data Subscription and Debug Display using Node-RED

Objective:
To subscribe DHT22 temperature data published by ESP32 using MQTT and display it in Node-RED Debug window.

Tools / Software Used:

- ESP32 (Simulator in Wokwi / VS Code)
- DHT22 sensor
- Mosquitto MQTT Broker (running on port 1883)
- Node-RED (http://localhost:1880)

Procedure (Step-by-Step):

1. **Start Mosquitto Broker:**
   Open terminal / command prompt and run:
2. mosquitto -v

   Output should confirm:

   mosquitto running

3. **Run ESP32 Code (Publisher):**
   - Make sure ESP32 code has correct broker IP:
   - const char* mqtt_server = "";
   - ESP32 publishes temperature to topic:
   - home/lab1/temp
   - Serial monitor shows:
   - MQTT connected
   - Publishing temperature: 26.5
4. **Start Node-RED:**

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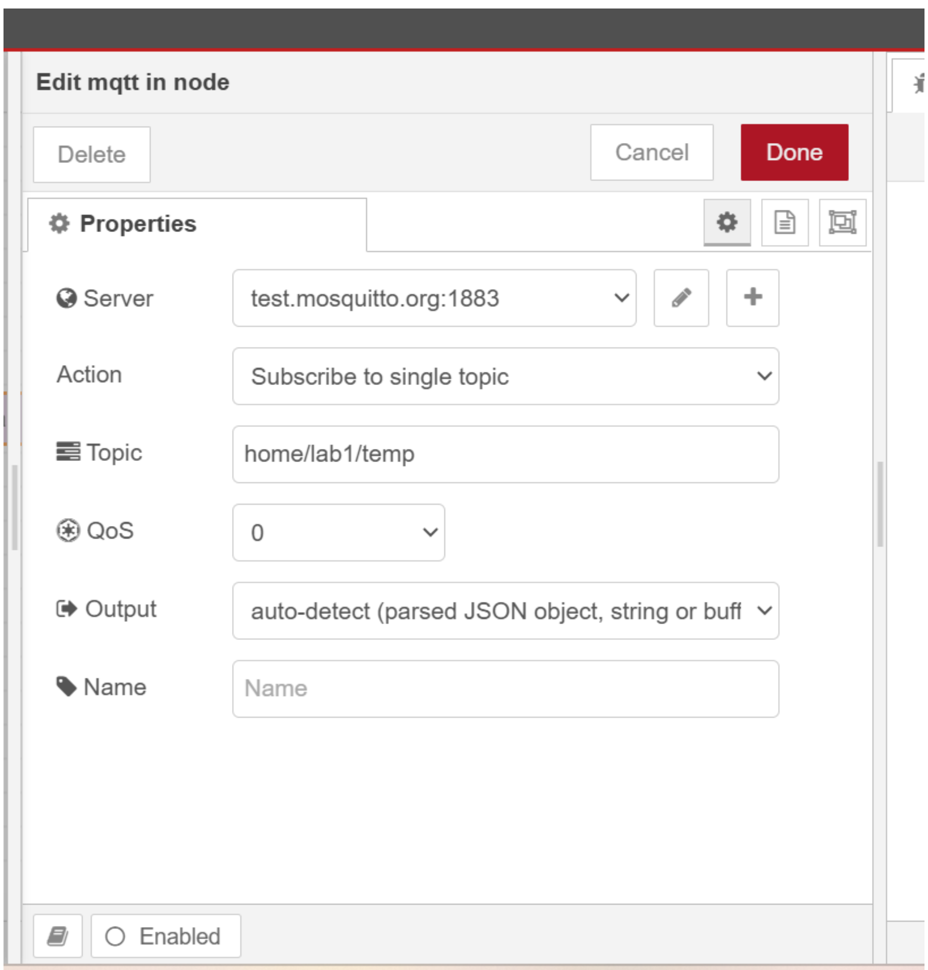
5. node-red

Open browser: http://localhost:1880

6. **Add MQTT-in Node:**
   - Drag **MQTT-in node**
   - Configure:
     - Server: localhost
     - Port: 1883
     - Topic: home/lab1/temp
     - QoS: 0
     - Action: Subscribe to single topic
   - Click **Done**



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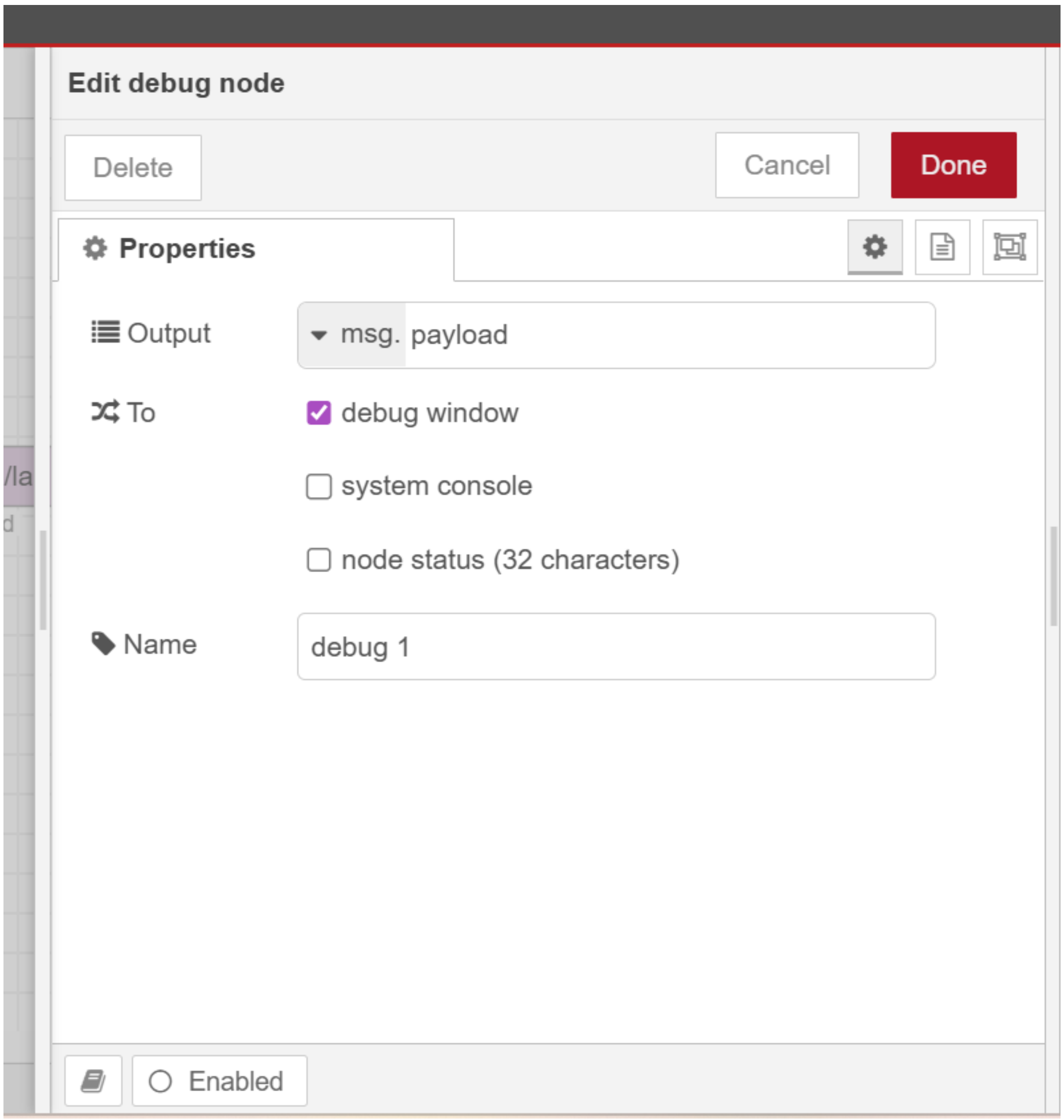
7. **Add Debug Node:**
   - Drag **Debug node**
   - Output: msg.payload
   - Connect: MQTT-in → Debug



8. **Deploy Flow:**
   - Click **Deploy**
   - Open **Debug tab** (right panel)
   - You should see temperature values like:
   - 26.4

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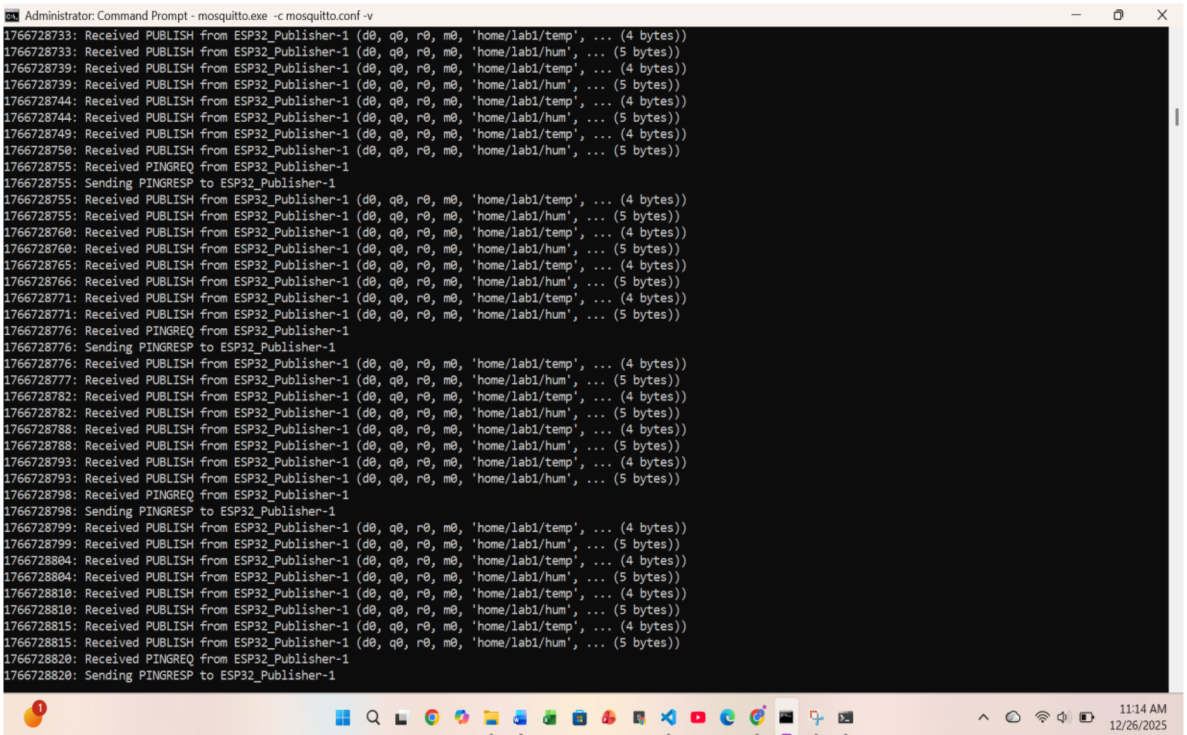
  - 27.0

9. **Screenshots:**

  - Mosquitto running terminal



  - ESP32 serial output (Wokwi)

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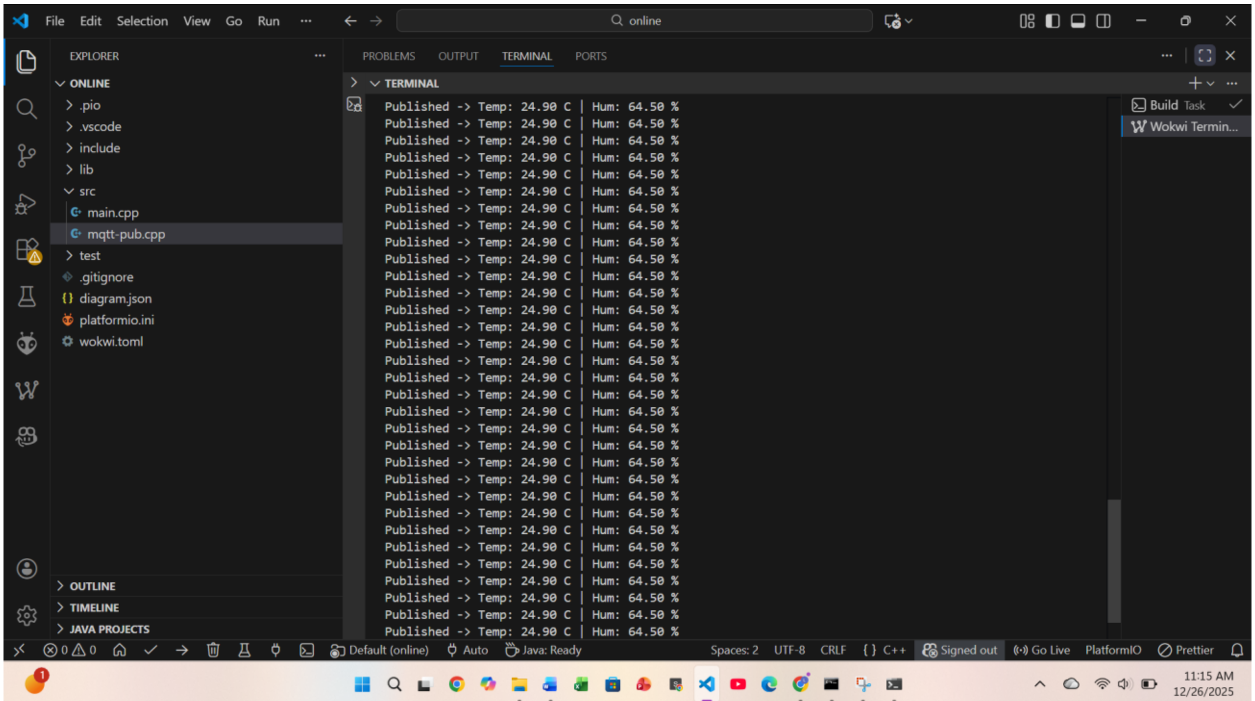
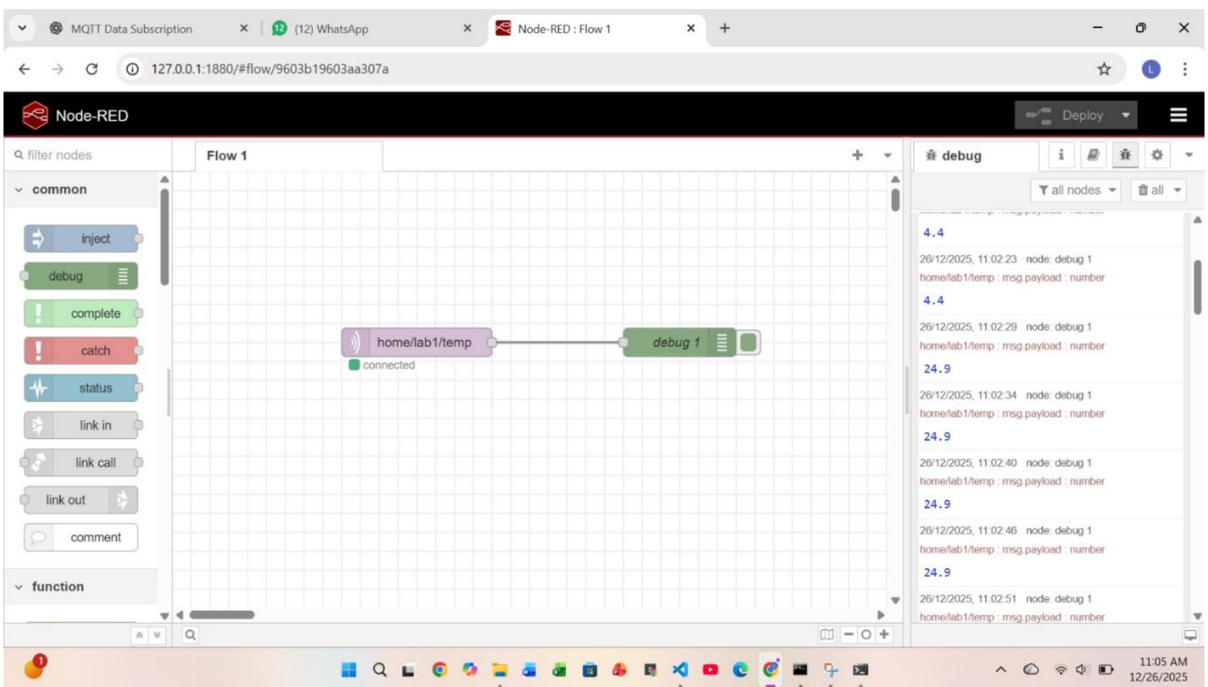


- Node-RED flow (MQTT-in → Debug)



Conclusion:
ESP32 successfully published DHT22 temperature data using MQTT protocol. Node-RED subscribed to the topic home/lab1/temp through a local Mosquitto broker. The received data was displayed in the Debug window, demonstrating successful IoT data monitoring.